6.3.4 Standard Deviation

Consider the following data sets.

10, 7, 6, 5, 4, 3, 2
10, 10, 10, 9, 9, 9, 2 ,2
10, 4, 4, 3, 2, 2, 2

It is obvious that the range for the three sets of data is 8. But a careful look at these sets clearly shows the numbers are different and there is a necessity for a new measure to address the real variations among the numbers in the three data sets. This variation is measured by standard deviation. The idea of standard deviation was given by Karl Pearson in 1893.

Definition:

'Standard deviation is the positive square root of average of the deviations of all the observation taken from the mean.' It is denoted by a greek letter σ.

(a) Ungrouped data

$x_1, x_2, x_3 \ldots x_n$ are the ungrouped data then standard deviation is calculated by

1. **Actual mean method:** Standard deviation $\sigma = \sqrt{\frac{\sum d^2}{n}}$, $d = x - \bar{x}$

2. **Assumed mean method:** Standard deviation $\sigma = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2}$, $d = x - A$

(b) Grouped Data (Discrete)

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \quad , d = x - A$$

Where, f = frequency of each class interval

N = total number of observation (or elements) in the population

x = mid – value of each class interval

where A is an assumed A.M.

(c) Grouped Data (continuous)

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times C, \quad d = \frac{x - A}{c}$$

Where, f = frequency of each class interval

N = total number of observation (or elements) in the population

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